

1. (80 points) Using Tables 3-1 and 3-2, or by direct calculation, determine the following:

- a. The Laplace transform of $(t + 6) * (u(t) - u(t - 4))$
- b. The inverse Laplace transform of $\frac{e^s}{s^3}$
- c. The Laplace transform of $(t + 4)e^t u(t)$
- d. The Laplace transform of $(t + 1)e^{t+1} u(t + 2)$
- e. The inverse Laplace transform of $\frac{1}{s^2 - 9}$
- f. The Laplace transform of $e^{-3t} \cos(2t + 125)$
- g. The Laplace transform of $\frac{d(t^3 e^{-t} u(t))}{dt}$
- h. The Laplace transform of $\frac{d^3 x(-t)}{dt^3}$, for an anticausal signal $x(-t)$
- i. The Laplace transform of $r(t + 2)$, where $r(t)$ represents the unit ramp signal
- j. The Laplace transform of $(t + 1)^2 e^{-3t} \cos(2t) u(t)$

2. (10 points) Calculate the initial and final values for the system $\mathbf{H}(s) = \frac{3s+2}{s^2+2s+5}$

3. (10 points) Calculate the initial and final values for the system $\mathbf{H}(s) = \frac{2s+1}{s^2-1}$

Suggestion: We have posted a collection of practice problems with solutions on the Canvas site to help you learn this topic better.